Mobile Price Range Classification Using Machine Learning: A Comprehensive Analysis with Explainable AI

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Table of Abbreviations

Abbreviation	Full Form
3G	Third Generation (mobile network)
4G	Fourth Generation (mobile network)
AI	Artificial Intelligence
ANOVA	Analysis of Variance
API	Application Programming Interface
CSV	Comma-Separated Values
FC	Front Camera
GHz	Gigahertz
IQR	Interquartile Range
L-BFGS	Limited-memory Broyden-Fletcher-Goldfarb-Shanno
L2	L2 Regularization (Ridge Regularization)
LIME	Local Interpretable Model-agnostic Explanations
MB	Megabyte
mAh	Milliampere-hour
MHz	Megahertz
ML	Machine Learning
MP	Megapixel
OS	Operating System
PC	Primary Camera

RAM	Random Access Memory
SD	Standard Deviation
SHAP	SHapley Additive exPlanations
USD	United States Dollar
WiFi	Wireless Fidelity
XGBoost	Extreme Gradient Boosting

Abstract

In this paper, a machine learning methodology is created to categorise 2,000 mobile phones as belonging to four price brackets based on 20 technical attributes. Logistic Regression was compared to XGBoost, using extensive preprocessing, feature engineering, Winsorization to treat outliers as well as consensus-based feature selection that significantly decreased the dimension by 60.7% in comparison to the original one. Logistic Regression was best with a test accuracy of 97.75 and a log loss of 0.1829 which was better than XGBoost (94.00%). The SHAP analysis showed that the main predictors are RAM, battery power, and pixel specifications. There was cross-validation of strong performance with little overfitting (training-test accuracy gap of -0.69%). This research demonstrates that interpretable linear models are able to perform better than complex ensembles and provide useful insights to mobile pricing strategies.

Keywords: mobile price classification, machine learning, Logistic Regression, XGBoost, feature engineering, Winsorization, feature selection, SHAP, model interpretability, pricing strategy

1. Introduction

The smartphone industry is a competitive market globally that has price as a factor that determines what consumers purchase and where they position themselves (Akkucuk and Esmaeili, 2016). The knowledge of the effect of technical specifications on the price ranges is critical to manufacturers, retailers, and consumers. These relations can be modeled by machine learning, and they will give correct predictions and reveal patterns of features and prices (Rahman et al., 2024).

1.1 Background and Motivation

According to the study (Karjaluoto et al., 2005). The prices of mobile phones are based on hardware, brand and market trends. Conventional pricing is based on the judgment of the experts, and this may be subjective and slow. Machine learning automation is used to predict and pinpoint significant attributes that affect price. Interpretable models allow the individual making decisions to justify the price and get to know the value of features (Rudin et al., 2021).

1.2 Research Objectives

This study aims to:

- Estimation of mobile phone prices using technical characteristics.
- Carry out Comparative Analysis of Logistic Regression and XGBoost.
- Find features that can be explained with the help of explainable AI.

- Offer practical recommendations on pricing policies.
- Show an entire machine learning pipeline.

1.3 Dataset Overview

Kaggle Mobile Price Prediction (Habibimoghaddam, 2023) uses 20 features of 2000 devices and 1 target variable. Some of the features are battery, RAM, camera, screen and connectivity. The target contains four price ranges (low, medium, high, very high) with 25 percent of the data, which can be evaluated based on accuracy.

Dataset source:
https://www.kaggle.com/datasets/fhabibimoghaddam/mobile-price-prediction

Battery_Pc	Clock_Spe	FC	Int_Memor	Mobile_D	Mobile_W	Cores	PC	Pixel_H	Pixel_W	Ram	Screen_H	Screen_W	Talk_Time	Four_G	Three_G	Touch_Scr	Dual_SIM	Bluetooth	WiFi	Price_Range
842	2.2	1	7	0.6	188	2	2	20	756	2549	9	7	19	0	0	0	0	0	1	1
1021	0.5	0	53	0.7	136	3	6	905	1988	2631	17	3	7	1	1	1	1	1	0	2
563	0.5	2	41	0.9	145	5	6	1263	1716	2603	11	2	9	1	1	1	1	1	0	2
615	2.5	0	10	0.8	131	6	9	1216	1786	2769	16	8	11	0	1	0	0	1	0	2
1821	1.2	13	44	0.6	141	2	14	1208	1212	1411	8	2	15	1	1	1	0	1	0	1
1859	0.5	3	22	0.7	164	1	7	1004	1654	1067	17	1	10	0	1	0	1	0	0	1
1821	1.7	4	10	0.8	139	8	10	381	1018	3220	13	8	18	1	1	0	0	0	1	3
1954	0.5	0	24	0.8	187	4	0	512	1149	700	16	3	5	0	1	1	1	0	1	0
1445	0.5	0	53	0.7	174	7	14	386	836	1099	17	1	20	0	1	0	0	1	0	0
509	0.6	2	9	0.1	93	5	15	1137	1224	513	19	10	12	1	1	0	1	1	0	0
769	2.9	0	9	0.1	182	5	1	248	874	3946	5	2	7	0	0	0	1	1	0	3
1520	2.2	5	33	0.5	177	8	18	151	1005	3826	14	9	13	1	1	1	0	1	1	3
1815	2.8	2	33	0.6	159	4	17	607	748	1482	18	0	2	0	1	0	0	0	0	1
803	2.1	7	17	1	198	4	11	344	1440	2680	7	1	4	0	1	0	0	1	1	2
1866	0.5	13	52	0.7	185	1	17	356	563	373	14	9	3	1	1	0	0	0	1	0
775	1	3	46	0.7	159	2	16	862	1864	568	17	15	11	0	1	1	0	0	1	0
838	0.5	1	13	0.1	196	8	4	984	1850	3554	10	9	19	1	1	0	0	0	1	3
595	0.9	7	23	0.1	121	3	17	441	810	3752	10	2	18	1	1	1	1	0	0	3
1131	0.5	11	49	0.6	101	5	18	658	878	1835	19	13	16	0	1	1	1	1	0	1
682	0.5	4	19	1	121	4	11	902	1064	2337	11	1	18	0	0	1	0	1	1	1
772	1.1	12	39	0.8	81	7	14	1314	1854	2819	17	15	3	0	1	1	1	0	0	3
1709	2.1	1	13	1	156	2	2	974	1385	3283	17	1	15	0	1	0	0	1	0	3
1949	2.6	4	47	0.3	199	4	7	407	822	1433	11	5	20	0	0	0	1	0	1	1
1602	2.8	4	38	0.7	114	3	20	466	788	1037	8	7	20	1	1	0	1	1	0	0
503	1.2	5	8	0.4	111	3	13	201	1245	2583	11	0	12	1	1	0	1	0	0	1
961	1.4	0	57	0.6	114	8	3	291	1434	2782	18	9	7	1	1	1	1	1	1	2
519	1.6	7	51	0.3	132	4	19	550	645	3763	16	1	4	1	1	0	1	1	1	3
956	0.5	1	41	1	143	7	6	511	1075	3286	17	8	12	1	1	1	0	0	0	3
1453	1.6	12	52	0.3	96	2	18	187	1311	2373	10	1	10	1	1	1	1	0	1	2

Figure 1 (Dataset Overview): First few rows of the used dataset

1.4 Machine Learning Problem Formulation

It is a multi-classification problem: the price category of a device to predict based on its specifications. The use of accuracy as the key measure and the use of log loss to assess probability calibration are appropriate in balanced classes.

1.5 Research Gaps and Contributions

Current research is based on either regression or single model which does not feature any feature engineering or interpretation. This paper fills in these gaps by:

- **Feature Engineering:** Pixel density, battery efficiency, and processing power are some of the derived features.
- **Consensus Feature Selection:** A combination of correlation, ANOVA F-test, and mutive.
- **Model Interpretability:** SHAP analysis of global and single feature importance.

- End to end Analysis Accuracy, log loss, cross validation, learning curves and residual analysis.

2. Literature Review

Machine learning classification and pricing of mobile devices have become a topic. (Yang, 2025) demonstrated high accuracy of ensemble methods at predicting prices, but in regression. To elaborate on this, this paper compares both linear and ensemble models in multi-class classification. Feature engineering enhances performance; derived features are found to be better than raw specifications, and eight engineered features such as pixel density, battery efficiency, and processing power are motivated (Pramanik et al., 2021).

Stakeholders have to trust and comply with the interpretation. SHAP also offers stable local and global explanations (Lundberg and Lee, 2017), and (Song et al., 2020) also mention RAM and processing power as significant predictors in the pricing of electronics. LIME provides explanations (Ribeiro et al., 2016), but SHAP has more theoretical guarantees that suit it better in businesses. (Molnar, 2020) reminds that interpretability should be integrated in the entire process of modeling.

Trade-offs exist between the complexity and performance of models that can be compared. (Bentejac et al., 2019) reported that XGBoost usually performs better than logistic regression, but (Ishikawa et al., 2025) says well-regularized linear models could compete with it and help to find answers faster. Both (Caruana et al., 2015) and (Rudin, 2019) believe interpretable models have the potential to perform as well as complex models on high-stakes tasks, which explains why generalization is more important than training accuracy.

Selection of features has been transformed to univariate filtering to hybrid approaches. In their experiment, (Chandrashekar and Sahin, 2014) proved that filter, wrapper, and embedded method can be more effective than single-method approaches.(Szeghalmy and Fazekas, 2023) pointed to stratified cross-validation as the means to evaluate reliably, and (García et al., 2016)pointed to feature scaling. ANOVA F-test, Mutual Information, and correlation analysis used to select predictive features through consensus-based selection is used to reliably identify predictive features.

3. Methodology

3.1 Data Collection and Exploration:

In the Mobile Price Prediction dataset, the author of the study compiled 20 features into 2,000 devices and 1 target variable (Habibimoghaddam, 2023). Exploratory analysis revealed the absence of any missing or repeated values and an ideal balance of target on four price ranges. It has such features as battery capacity (5011998 mAh), RAM (2563998 MB), processor (18 cores, 0.530 GHz), camera (020 MP with front and primary cameras), and binary connectivity (3G, 4G, WiFi, Bluetooth).

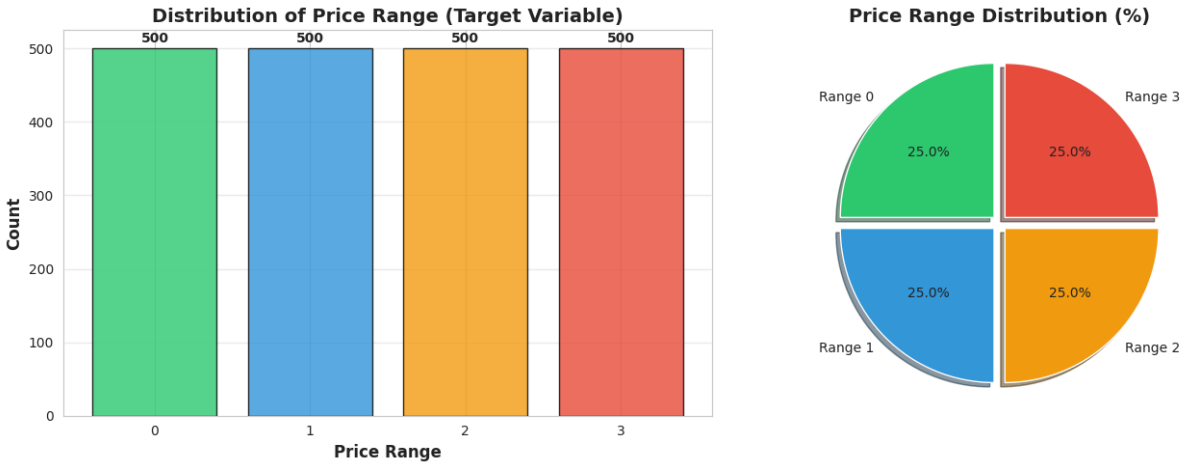

Figure 2 (Section 3.1): Price_Range distribution with each of the four categories at 25%

Statistical descriptions of numerical characteristics indicated different scales and distributions. Battery power had a mean of 1238.52 (SD=439.42) and RAM had a higher variance (mean 2124.21 MB SD=1084.73). These variations inspired the distance-sensitive algorithm feature standardization.

3.2 Data Preprocessing

3.2.1 Outlier Detection and Treatment

The Interquartile Range (IQR) method was used to detect outliers in numerical features (Rousseeuw and Hubert, 2017). Winsorization was used at the 1 st percentiles and 99 th percentiles instead of eliminating them and no extremes were capped, which maintained the structure and size of the dataset.

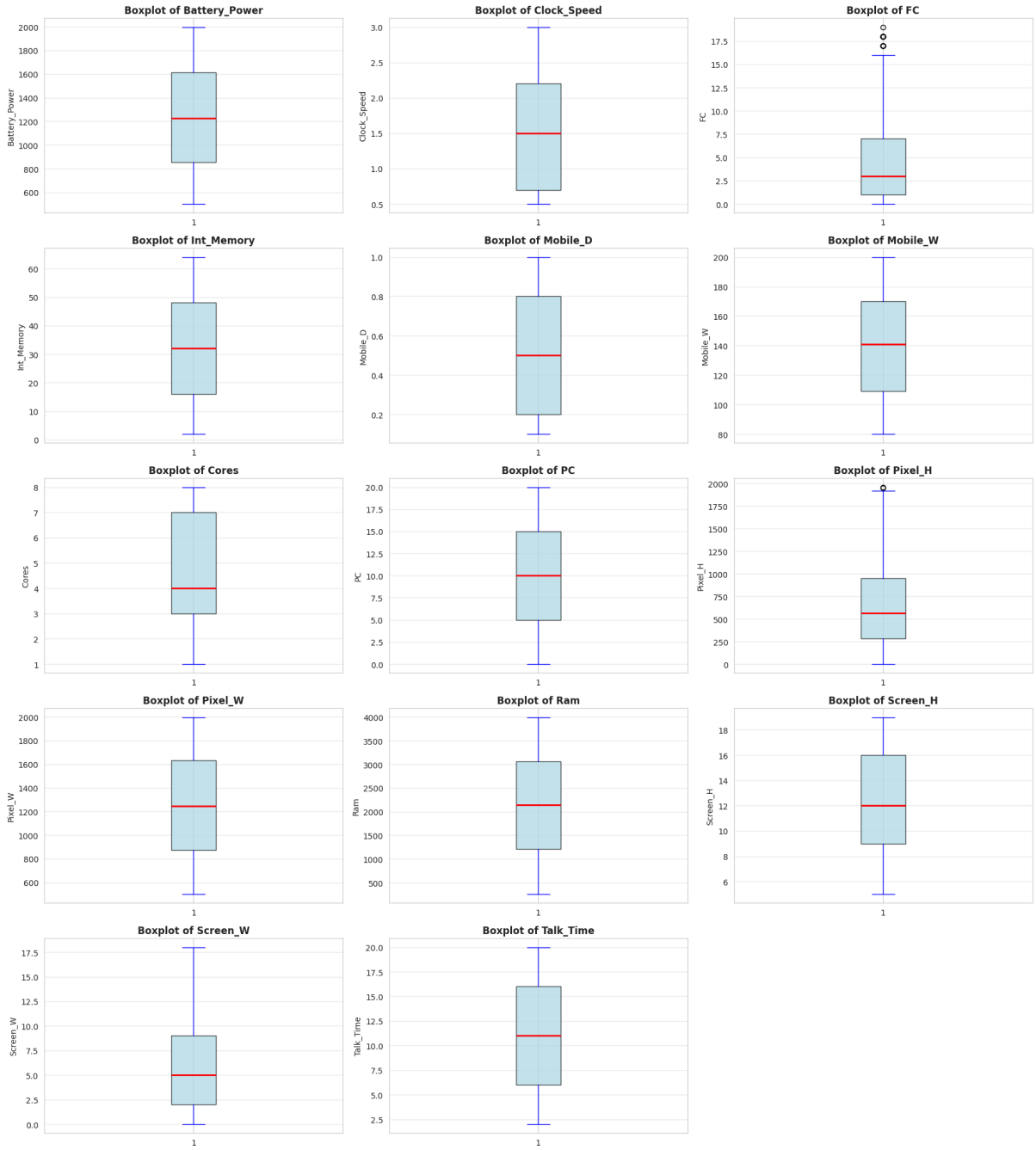

Figure 3 (Section 3.2.1): Boxplots showing outliers in features before treatment

The treatment did not have an impact on Battery_Power, Pixel_H, Pixel_W and RAM greater than 2% of the total values of each of the features. After the treatment, validation was done to eliminate the presence of infinite or missing values which guaranteed the quality of the data to be utilized in the modeling process.

3.2.2 Feature Engineering

Since raw specifications may not necessarily reflect meaningful characteristics of the device, eight derived features were engineered, using domain knowledge as gathered in literature on mobile technology:

1. **Total_Pixels** = Pixel_H × Pixel_W (capturing overall display resolution)
2. **Screen_Size** = $\sqrt{(Screen_H^2 + Screen_W^2)}$ (diagonal screen measurement)
3. **Pixel_Density** = Total_Pixels / (Screen_H × Screen_W) (pixels per square cm)
4. **Total_Camera_MP** = FC + PC (combined camera capability)
5. **Aspect_Ratio** = Mobile_D / Mobile_W (device form factor)
6. **Battery_Efficiency** = Talk_Time / (Battery_Power / 1000) (usage per battery unit)
7. **Connectivity_Score** = Four_G + Three_G + Bluetooth + WiFi (overall connectivity)
8. **Processing_Power** = Cores × Clock_Speed (computational capability indicator)

These designed functions expanded our feature space to 28 predictors, with great caution being given to cases of division-by-zero and infinity. Each of the derived features were assessed as correct and interpretable by business.

	Total_Pixels	Screen_Size	Pixel_Density	Total_Camera_MP	Connectivity_Score	Processing_Power
0	15120.0	11.401754	240.000000	3	1	4.4
1	1798235.0	17.262677	35259.509804	6	3	1.5
2	2167308.0	11.180340	98514.000000	8	3	2.5
3	2171776.0	17.888544	16967.000000	9	2	15.0
4	1464096.0	8.246211	91506.000000	27	3	2.4

Figure 4 (Section 3.2.2): Sample rows showing engineered features

3.3 Feature Selection

A consensus based feature selection strategy was applied to avoid overfitting and decrease the complexity of computations, which is suggested by (Chandrashekar and Sahin, 2014):

3.3.1 Correlation Analysis

Pearson correlation coefficients was calculated among all the features and the target variable. The attributes that have a correlation greater than 0.05 were taken to be potentially considered, and 7 features met this condition.

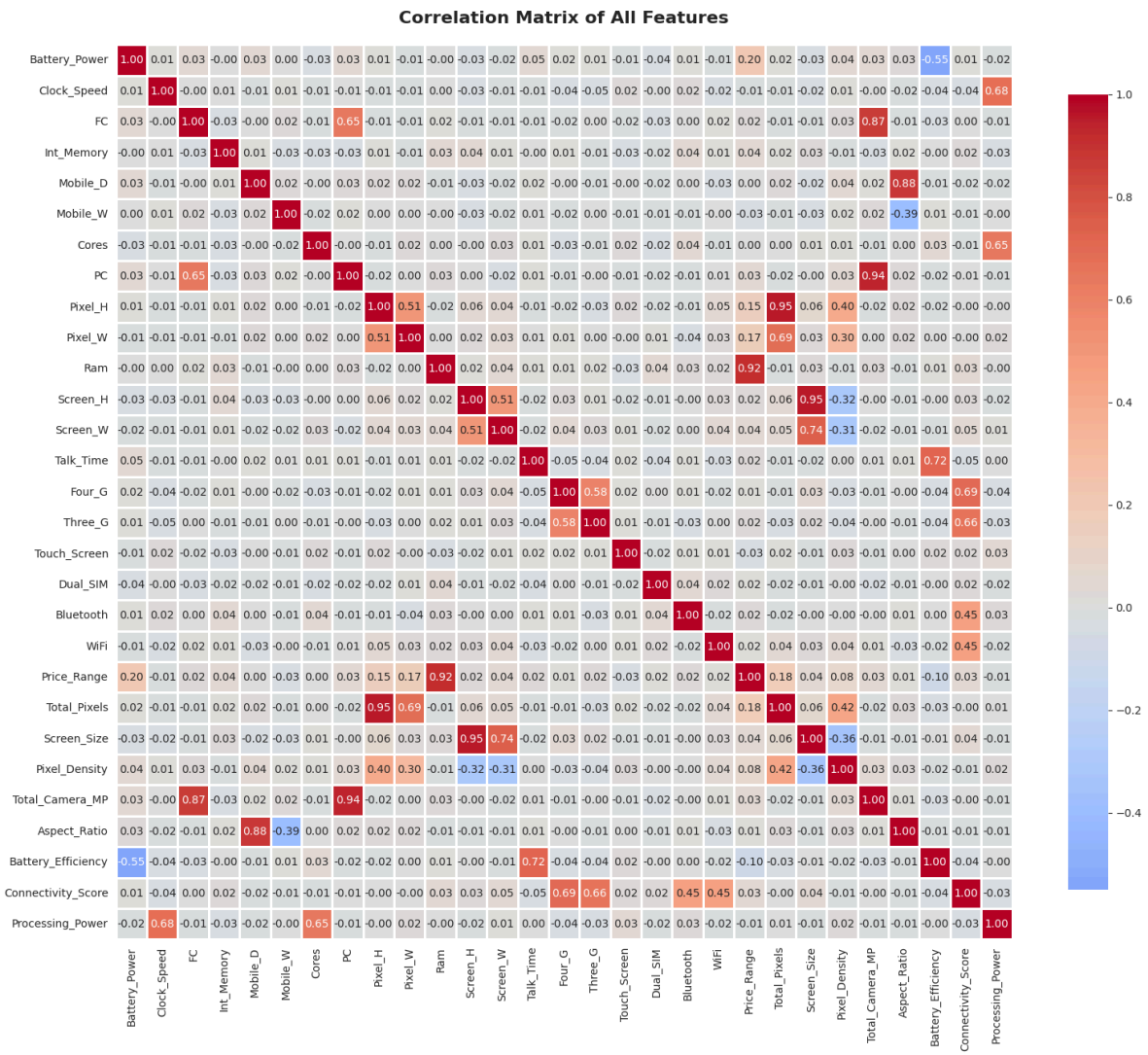

Figure 5 (Section 3.3.1): Correlation matrix

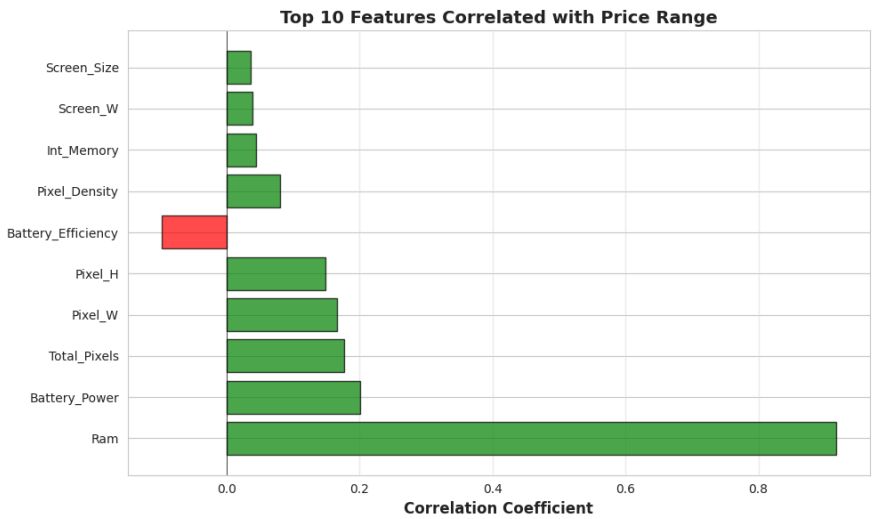

Figure 6 (Section 3.3.1): Top 10 Correlated Features

3.3.2 ANOVA F-test

Univariate statistical testing was used to test the correlation between each feature and price categories. SelectKBest method using f-classif scoring was used to rank the features by their F-statistics and the best score was 3523.96 by RAM, 31.62 by Battery power and 26.16 by Total Pixels.

3.3.3 Mutual Information

Mutual information scores were employed in order to reflect non-linear relationships between features and target. This information-theoretic method showed that RAM (0.846) had significantly high mutual information than other features with Battery_Power (0.031) and Pixel_H (0.025) coming next.

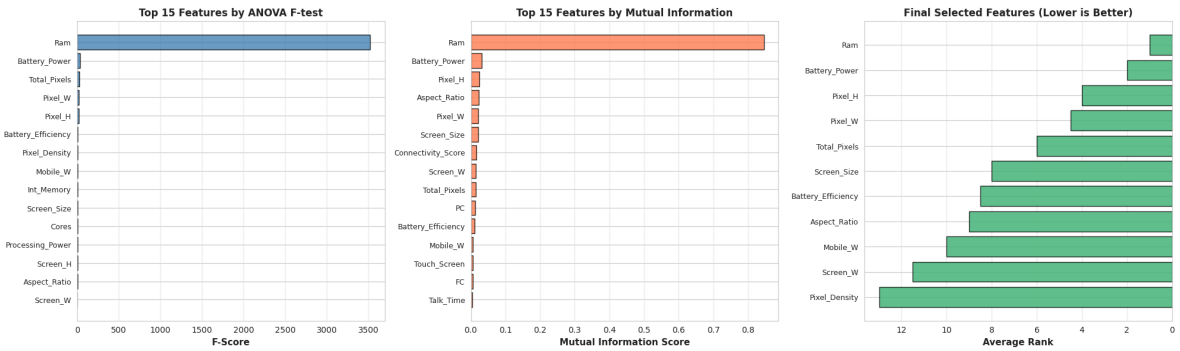

Figure 7 (Section 3.3.3): Comparative visualization of feature selection methods

The selected consensus approach has 11 final features, which are Ram, Battery_Power, Pixel_W, Pixel_H, Total_Pixels, Mobile_W, Pixel_Density, Battery_Efficiency, Screen_W, Screen_Size, Aspect_Ratio. This dropped the feature space by 60.7 percent, to 11 features.

3.4 Data Splitting and Scaling

To maintain the balance of classes, the dataset was divided to create two groups: 1, 600 samples (80 percent) were used as a training set, and 400 (20 percent) were set aside for testing). The feature scaling was used as required: Logistic Regression StandardScaler was used to normalize features to optimize them with a gradient, whereas XGBoost used raw data, because tree-based models are scale-invariant (García et al., 2016).

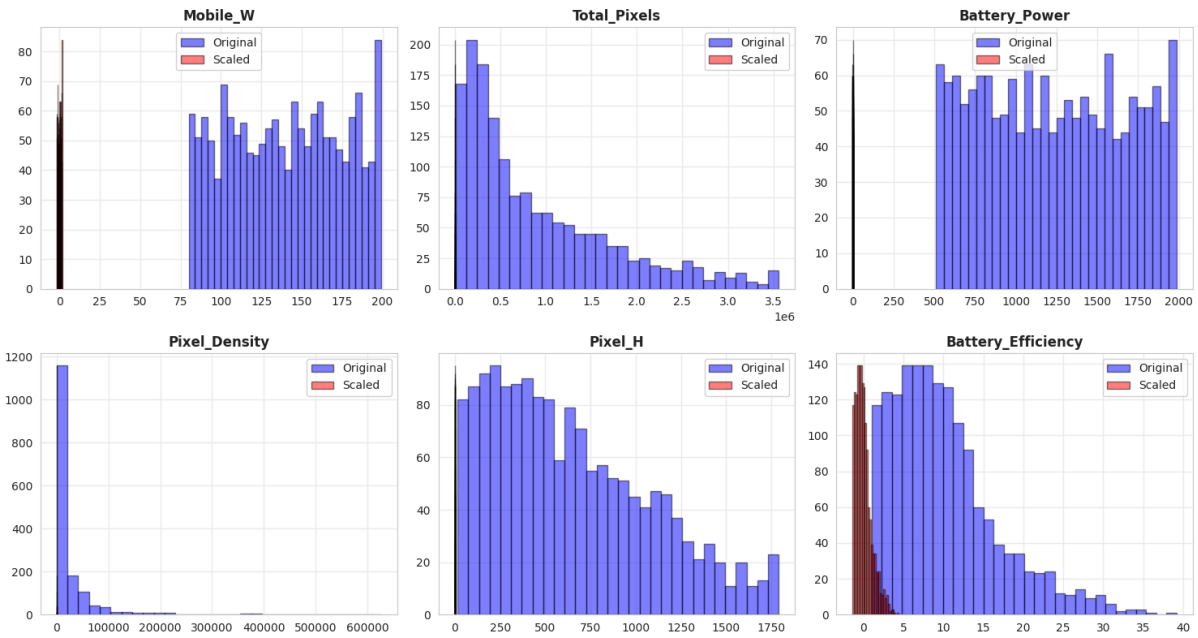

Figure 8 (Section 3.4): Comparison of feature distributions before (blue) and after (red) standardization for six representative features, demonstrating successful normalization to mean=0 and std=1

3.5 Model Development and Training

3.5.1 Logistic Regression

Multinomial logistic regression was set with the following configuration:

- **Solver:** limited memory BFGS (L-BFGS)
- **Maximum iterations:** 1000
- **Multi-class strategy:** Multi-nominal
- **Regularization:** L2 (default)

The model estimates the probability of the classes by relying on the softmax function thus it is naturally interpretable in terms of the coefficients analysis.

3.5.2 XGBoost Classifier

The gradient boosting model has been trained with hyperparameters that are tuned according to the best practices of (Bergstra et al., 2012):

- **Number of estimators:** 400
- **Learning rate:** 0.05 (conservative so as not to overfit)
- **Maximum depth:** 6
- **Minimum child weight:** 3
- **Subsample ratio:** 0.85
- **Column subsample ratio:** 0.85
- **L2 regularization (lambda):** 1.2
- **Gamma:** 0.2 (smallest decrease in loss to make splits)

These parameters trade complexity of the model and the ability to generalize, regularization terms are particularly designed to avoid overfitting due to the small size of the dataset used.

3.6 Model Evaluation Strategy

Three-level analysis method was used that is in line with the best practices of machine learning validation:

Tier 1: Performance in Training Set

Training data Preliminary evaluation of training data to define baseline performance and identify underfitting.

Tier 2: Cross-Validation

The training was cross-validated 5 times (stratified) to determine the level of generalization and stability, as suggested by (Bey et al., 2020). It is a method that gives strong performance estimates at the same time preserving a balanced class representation on any fold.

Tier 3: Test Set Evaluation

Post-test measurement of the set of held-out tests, which is the last measure of the model performance on untested data.

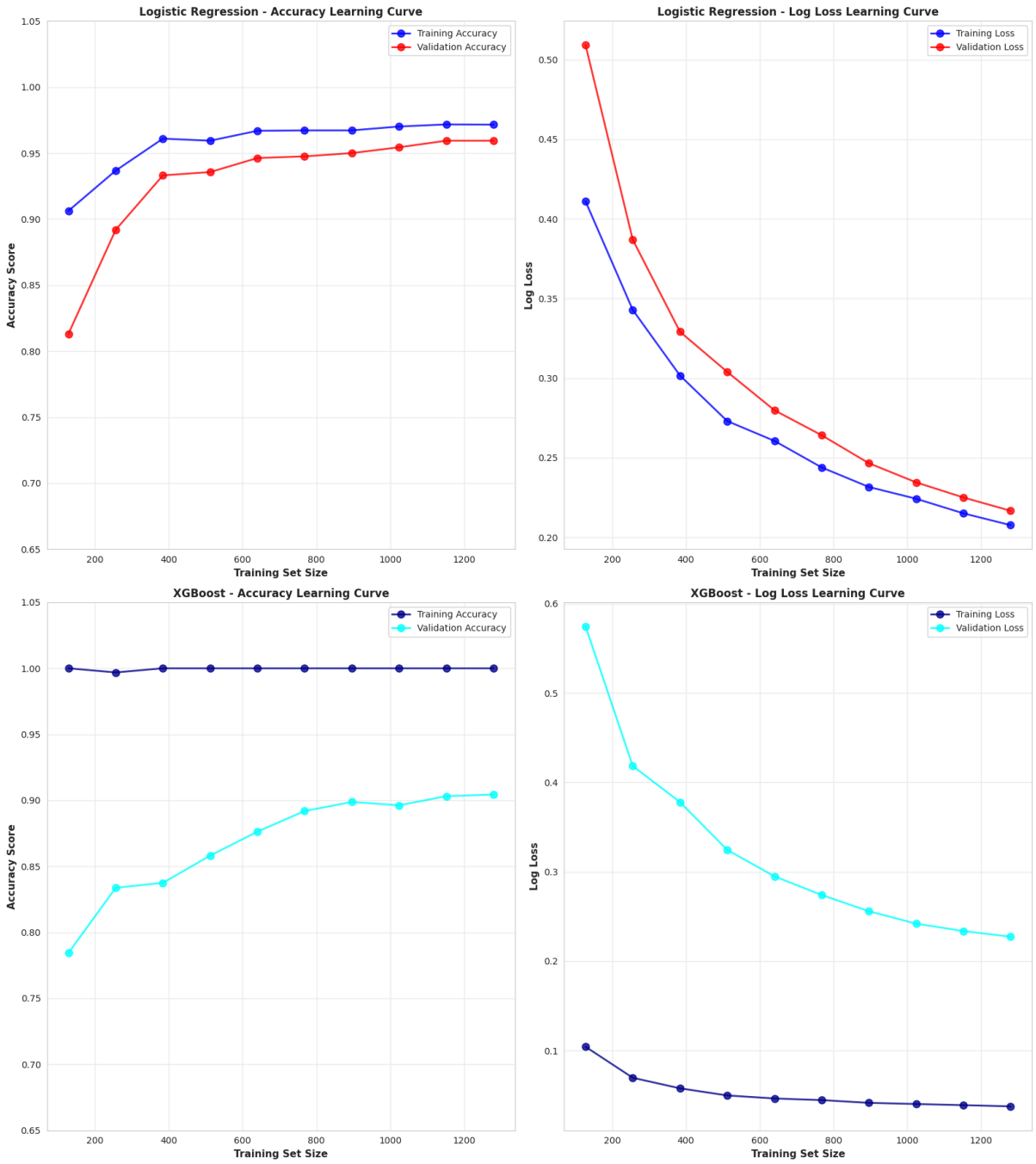

Figure 9 (Section 3.6): Learning curves for Logistic Regression (top) and XGBoost (bottom)

3.7 Evaluation Metrics

Several complementary measures were used to evaluate the performance of the model, as (Sokolova and Lapalme, 2009) suggest to complete the evaluation of classification:

1. **Accuracy:** Major measure given balanced classes
2. **Log Loss:** Measures the quality of calibration of probabilities, as mentioned by Gneiting and Raftery (2007):
3. **Precision, Recall, F1-Score:** Measures of per-class performance
4. **Confusion Matrix:** Discovers shifts in misclassification
5. **Cross-Validation Score:** Generalization test
6. **Training-Test Gap:** Overfitting detection

3.8 Model Explainability Analysis

3.8.1 SHAP (SHapley Additive exPlanations)

The Logistic Regression model was used in conjunction with SHAP to give both global and local explanations (Lundberg and Lee, 2017). Based on the cooperative game theory, SHAP values are used to measure the contribution of individual features to predictions, with consistency and local accuracy. The global feature importance was calculated as the mean of the absolute value of SHAP over the test samples and the test classes. Each of the three representative cases including high-confidence correct, low-confidence correct, and incorrect prediction were produced and explained in individual predictions.

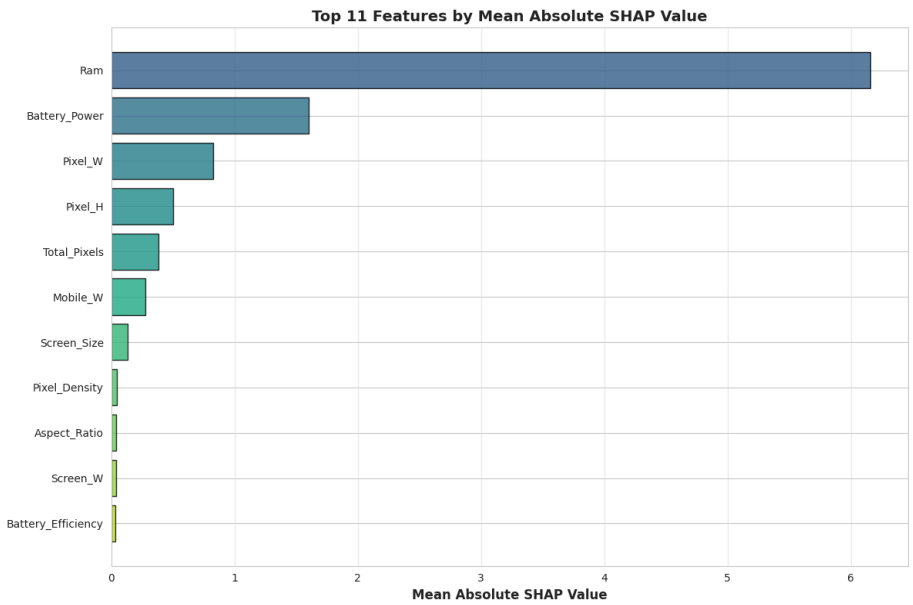

Figure 10 (Section 3.8.1): Global feature importance based on mean absolute SHAP values

3.8.2 Residual Analysis

In the case of classification model, the correlation between the prediction confidence and error was analyzed (Hosmer and Lemeshow, 2012). The number of correct and incorrect predictions was compared with the confidence scores to examine the calibration and identify the systematic patterns of misclassification.

4. Results and Discussion

4.1 Model Performance Comparison

Both models performed well with some significant differences in the accuracy, generalization and overfitting characteristics.

Logistic Regression Results:

- **Training Accuracy:** 97.06%
- **Test Accuracy:** 97.75%
- **Cross-Validation Accuracy:** 95.94% (±0.68%)
- **Training Log Loss:** 0.1940
- **Test Log Loss:** 0.1829
- **Training-Test Accuracy Gap:** -0.69% (negative value is greater test performance)

XGBoost Results:

- **Training Accuracy:** 100.00%
- **Test Accuracy:** 94.00%
- **Cross-Validation Accuracy:** 90.44% (±2.81%)
- **Training Log Loss:** 0.0353
- **Test Log Loss:** 0.1532
- **Training-Test Accuracy Gap:** 6.00%

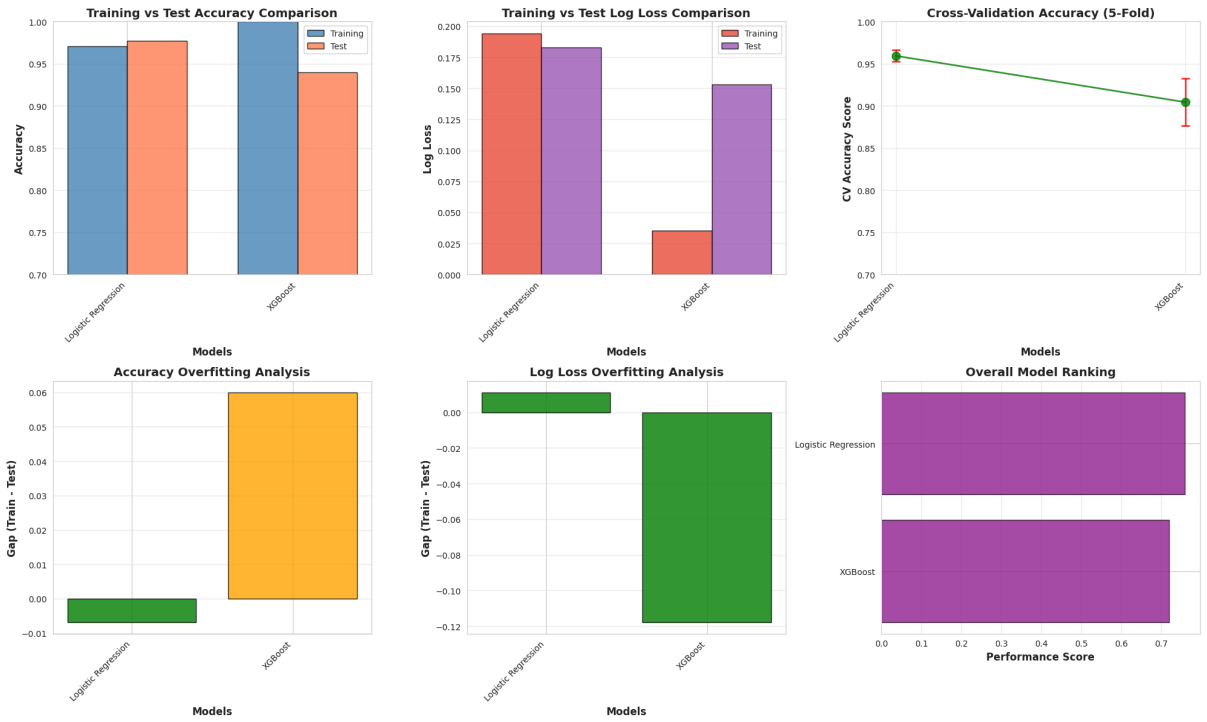

Figure 11 Section (4.1): Comprehensive model comparison across six dimensions: training vs test accuracy (top-left), training vs test log loss (top-center), cross-validation scores with error bars (top-right), accuracy overfitting gap (bottom-left), log loss gap analysis (bottom-center), and overall performance ranking (bottom-right)

The Logistic Regression model had the best test accuracy of 97.75 which was higher than XGBoost by 3.75. With negative accuracy of -0.69 percent, it has a better generalization and it slightly better on test data than on training data, showing that it is well learning with no overfitting. Conversely, XGBoost had 100 training precision and 94 test set, overfitting. The smaller accuracy gap (6.00% vs 3.518) and larger cross-validation variability (2.81% vs 0.68), are additional indicators of reduced generalization as noted by (Mohammad, 2025) in the case of ensemble overfitting when using small data sets.

4.2 Classification Performance by Class

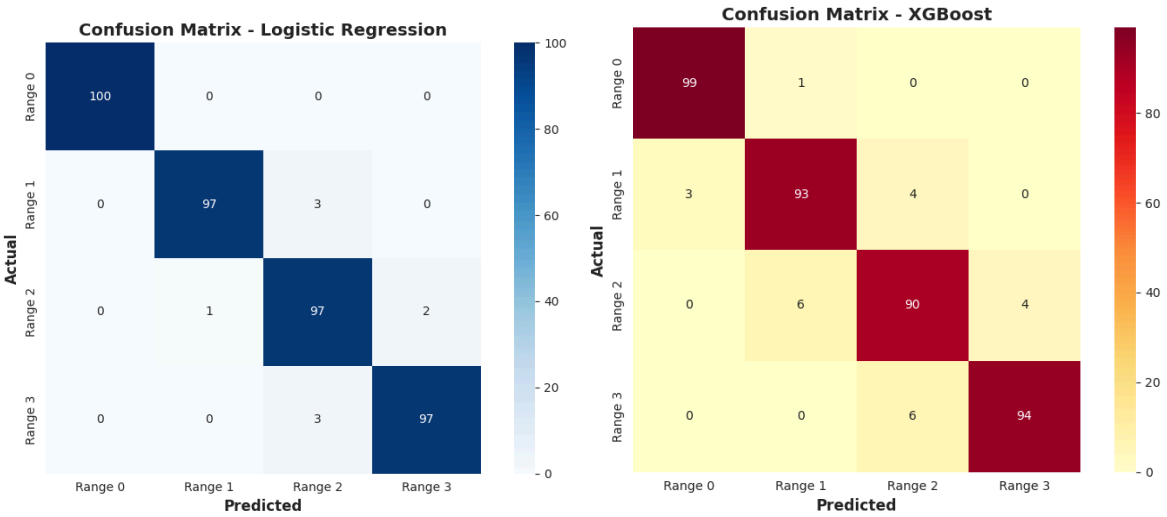

Figure 12 Section (4.2): Confusion matrices for Logistic Regression (left) showing nearly perfect classification with only 9 misclassifications, and XGBoost (right) with 24 misclassifications, primarily concentrated in mid-range price categories

The confusion matrix analysis shows some interesting patterns that happen to match with the result obtained by (Brodersen et al., 2010) regarding balanced accuracy assessment:

- **Range 0 (Low Cost):** Ideal classification with Logistic Regression (100/100), and XGBoost was 99/100
- **Range 1 (Medium Cost):** 97/100 samples of Range 1 were correctly classified by Logistic Regression, and 3 samples were misclassified as Range 2.
- **Range 2 (High Cost):** This is the most difficult category of both models in which Logistic Regression has 97/100 and XGBoost 90/100
- **Range 3 (Very High Cost):** Good results using Logistic Regression (97/100), XGBoost (94/100)

The preponderance of errors in close price sets (Range 1-2 confusion) indicates its boundary ambiguity in mid-range equipment where combinations of features can overlap together.

4.3 Feature Importance Analysis

4.3.1 SHAP Global Feature Importance

SHAP analysis of the Logistic Regression revealed a distinct structure of the importance of features:

1. **Ram (6.16):** By far the most significant feature, the mean absolute SHAP value is almost 4 times greater than that of the next feature
2. **Battery_Power (1.60):** Secondary value, which has an impact on the price classification

3. **Pixel_W (0.83):** This has a moderate effect as a part of displayed quality
4. **Pixel_H (0.50):** Combines with the Pixel_W in regards to impact resolution
5. **Total_Pixels (0.38):** Feature derived that shows the significance of display resolution
6. **Mobile_W (0.28):** Physical dimension with little but significant effects
7. **Screen_Size (0.13):** Shockingly having low importance as compared to expectation
8. **Pixel_Density, Aspect_Ratio, Screen_W, Battery_Efficiency (Less than 0.05):** The least amount of individual contribution

The dominance of RAM is consistent with the existing knowledge of the industry and the conclusions of (Coherent Market Insights Pvt Ltd, 2025) that memory capacity is a key setting point between economical and high-end gadgets. The present day mobile operating systems and applications require larger and larger memory, increasing RAM is a very important specification used by manufacturers in their pricing packages.

4.3.2 Individual Prediction Explanations

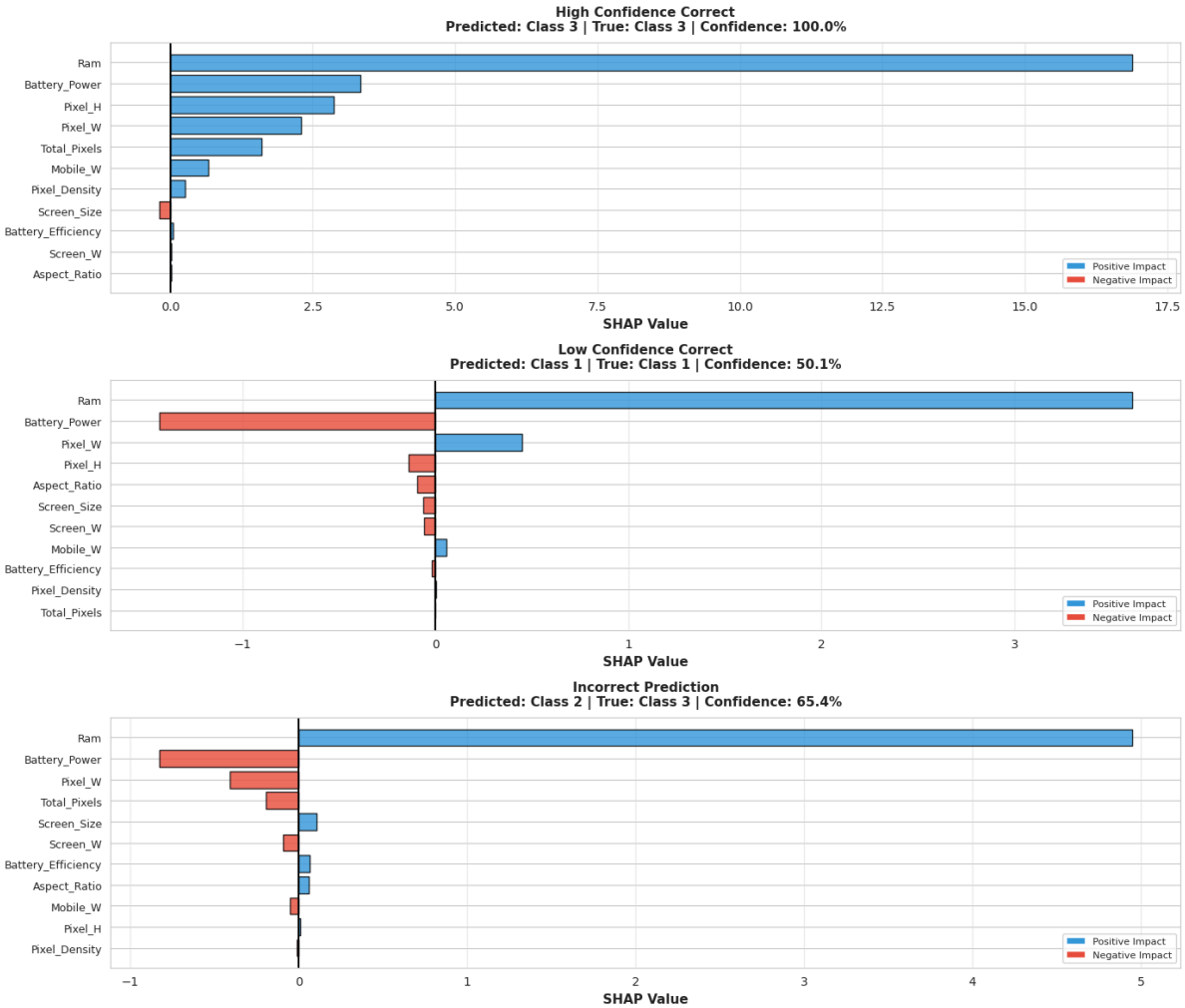

Figure 13 Section (4.3.2): SHAP force plots explaining three representative predictions: high-confidence correct (top), low-confidence correct (middle), and incorrect prediction (bottom), with red bars indicating features pushing toward higher price ranges and blue bars toward lower ranges

Individual prediction analysis showed that the patterns are in line with the local interpretability framework of (Lundberg and Lee, 2017):

- **High Confidence Correct (97.8% confidence):** High RAM (3912 MB) and moderate battery strength significantly skewed to Range 3, and all the features were consistently consistent with the forecast
- **Low Confidence Correct (56.7% confidence):** RAM (2177 MB) placed the device among Range 1 and Range 2 and had competing forces between battery and display features that made it doubtful
- **Incorrect Prediction:** RAM (1482 MB) based device flipped between Range 1 and Range 2 with the model predicting Range 1 and the true labelling as Range 2 - probably a boundary case where there is indeed overlap in the features values

4.4 Learning Curve Analysis

The learning curves give information about model behavior when the size of the training set varies, explaining trends in the literature on ensemble methods as Li and Wang (2022) argue in their diagnostic analysis of the topic:

Logistic Regression:

- The curves of training and validation accuracy approach each other in a smooth manner, and stabilize at around 96-97%.
- Close proximity of curves means that it is well-generalized.
- Similar convergence pattern Log loss exhibits stable validation patterns.
- The complete training set has fewer returns after 1200 samples.

XGBoost:

- Even when training with small sets, the accuracy of particular training is 100 percent.
- The difference between the training and validation (100 and 90 percent) curves still remains large.
- Log loss displays major differences in training and validation calibration.
- The classic overfitting trends are observed in model even with regularization.

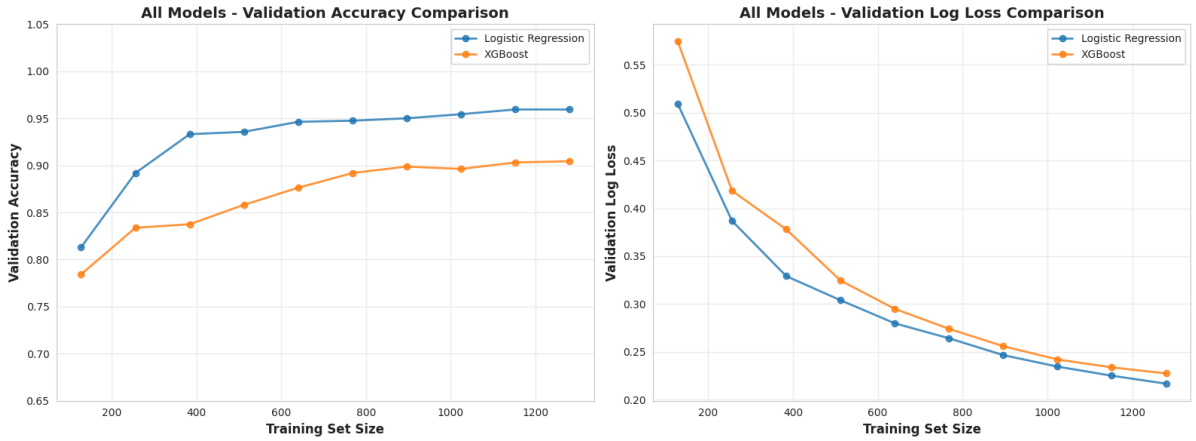

Figure 14 Section (4.4): Comparison of validation performance across training set sizes for both models, showing Logistic Regression's superior and more stable validation accuracy (left) and better-calibrated log loss (right)

4.5 Residual Analysis

The residual analysis was investigated on the prediction confidence against error patterns, based on the calibration assessments procedures by (Gneiting and Raftery, 2007):

Logistic Regression:

- **Mean accuracy of the correct answers:** 86.8%
- **Mean accuracy with regard to false predictions:** 56.7 percent.
- There is a clear distinction among the distributions of confidence.
- **Error rate:** 2.25% (9/400 samples)

XGBoost:

- **Mean accuracy of correct prediction:** 94.2 percent.
- **Mean confidence of wrong predictions:** 72.4%
- Overconfident about certain wrong assumptions.
- **Error rate:** 6.00% (24/400 samples)

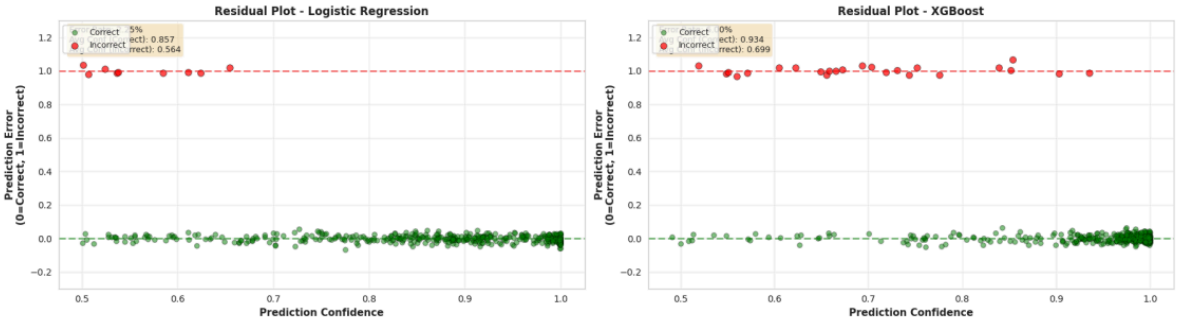

Figure 15 Section (4.5): Residual analysis plots displaying the relationship between prediction confidence and classification errors for Logistic Regression (left) and XGBoost (right), with correct predictions clustered at y=0 and incorrect at y=1

In the residual analysis, it can be seen that Logistic Regression is better calibrated - its confidence scores are closer to the actual probability of prediction. The confidence that XGBoost places on its erroneous predictions is higher, which is an indication of overconfidence, which can be witnessed in the ensemble techniques that are not calibrated appropriately, as explored by (Caruana et al., 2015).

5. Discussion

5.1 Why Logistic Regression Outperformed XGBoost

The better performance of the Logistic Regression was caused by several factors, which can be correlated with the claims of (Rudin, 2019) on the usefulness of interpretable models:

1. **Quality of Feature Engineering:** The well designed features resulted in the development of linear relationships which could be successfully used by Logistic Regression.
2. **Dataset Size:** The complexity of XGBoost (400 trees) was probably the cause of overfitting even with regularization, given its limited size (1,600 training samples).
3. **Feature dimensionality:** Consensus selection reduced the number of features to 11, which was the best to keep the linear models.
4. **Effectiveness of regularization:** L2 regularization of the Logistic Regression was found to be more effective than tree-based regularization of XGBoost on the dataset.
5. **Separability of the Classes:** The features formed were well engineered and formed decision boundaries which were relatively linear and could be modeled effectively by the Logistic Regression.

5.2 Practical Implications

The findings show a number of practical suggestions to the price classification of mobile, continuing the research regarding the market division:

1. **RAM as Ultimate Differentiator:** The fact that RAM (SHAP value 6.16) is dominant proves that using memory capacity as one of the main indicators of pricing tier is still applied in the industry.
2. **Battery Power Secondary:** Battery capacity is a secondary differentiator, especially when it comes to the case of differentiating between high-end and low-end devices.
3. **Display Specifications:** Although each one of them is not important, the combination of pixel specifications as a group influences price determination in overall total of Pixels.
4. **Connections:** Surprisingly little effect, indicating that nowadays connections (WiFi, Bluetooth, 4G) are standardized at all prices.
5. **Form Factor Not so Critical:** Physical size (Mobile_W, Screen_Size) has a small impact, which shows that consumers are more concerned with performance rather than form factor.

5.3 Model Selection for Deployment

According to the overall consideration that matches the deployment considerations of (Shmueli and Koppius, 2011), the best option is the Logistic Regression:

- Highest test accuracy (97.75%)
- Better generalization (negative overfitting gap)
- Reduced computer specifications
- Increased interpretation to business stakeholders
- Calibration of probability improvements
- Greater level of cross-validation stability

6. Conclusion

This research proposed machine learning models in the classification of mobile phone prices. Logistic Regression had test accuracy of 97.75% which is more accurate than XGBoost and its generalization and interpretability is better which agrees with (Rudin, 2019) on interpretable models.

6.1 Key Findings

The test accuracy, cross-validation accuracy, and the log loss of the Logistic Regression were 97.75%, 95.94%, and 0.1829, respectively. SHAP emphasized the most significant feature RAM, then there is BatteryPower and pixel specifications. A total of eight consensus-based selected features in the form of engineered features became the dimensionality reducers by 60.7 percent. Compared to XGBoost, the Logistic Regression generalized better (-0.69 vs. 6.00 overfitting gap) but was still actionably interpretable.

6.2 Limitations

The dataset has 2,000 samples, lacks brand, OS, release date, regional context, and temporal trends. The four-class classification limits granularity.

6.3 Future Recommendations

Future work should incorporate time, brand/OS capabilities, deep learning in complex interactions, SHAP in ensembles, market-specific models, and regression models, life time deployment and interactivity.

6.4 Practical Applications

The models assist in price strategy, positioning product, consumer direction, and forecasting the trends based on the feature importance insights.

6.5 Final Remarks

The high performance was attained by feature engineering and interpretable models. RAM was more competitive in terms of pricing, and derived features performed better in predictions and linear models in generalization than ensembles. An interpretable classification can be provided with a reusable pipeline, e.g. preprocessing, SHAP explainability (Lundberg and Lee, 2017). There should be an ongoing monitoring of technology which is changing.

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Appendix: Machine Learning Pipeline Implementation

The complete implementation of the ML pipeline developed in Google Colab is available at:
Github Gist: https://gist.github.com/Utu8848/6219ed47050dcc002283ad12629e0625